Evidence of nuclear geometry-driven anisotropic flow in OO and Ne-Ne collisions at $\sqrt{S_{NN}} = 5.36$ TeV

ALICE Collaboration

arXiv:2509.06428v1

Flow-like signals in small systems have been observed.

How can we be sure it is hydrodynamics?

This paper tests that by changing nucleus shape (O, Ne).

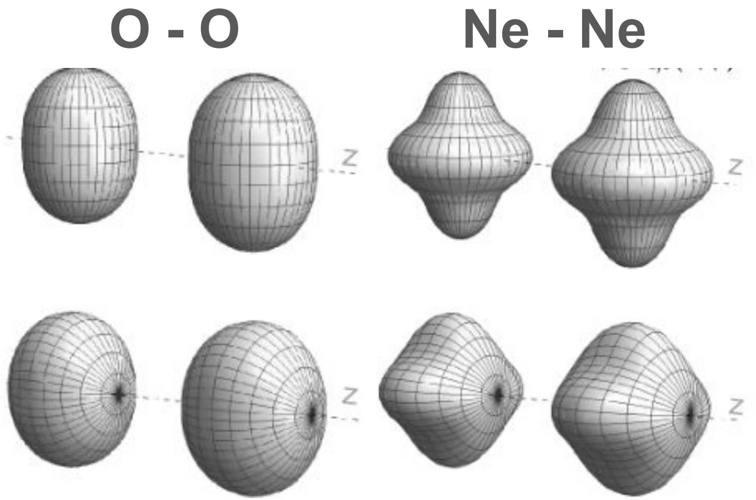
The Main Idea behind the study:

O-O Collisions v.s Ne-Ne Collisions

- Similar is size -> Similar bulk evolution
- OO is more symmetric
- NeNe is more deformed

Strategy:

- Calculate **ratio of flow observables**
  - Cancels many final state uncertainties
  - Isolates sensitivity to **initial geometry**.



https://journals.aps.org/prc/abstract/10.1103/PhysRevC.105.014905

Understanding Observables

Flow Coefficients: v_2 (Ellipitic flow) and v_3 (Triangular flow, Fluctuation)

$$\frac{dN}{d\varphi} \propto 1 + 2\sum_{n=1}^{\infty} v_n \cos[n(\varphi - \Psi_n)]$$

Cumulants: $v_2\{2\}$, $v_2\{4\}$, and $v_3\{2\}$

- Multiparticle correlations to measure flow
- Suppress non-flow effects

Probe system size and overlap geometry -> Centrality dependence

Calculate ratios to reduce uncertainties and isolate initial geometry

Baseline measurement

Quantify azimuthal anisotropy

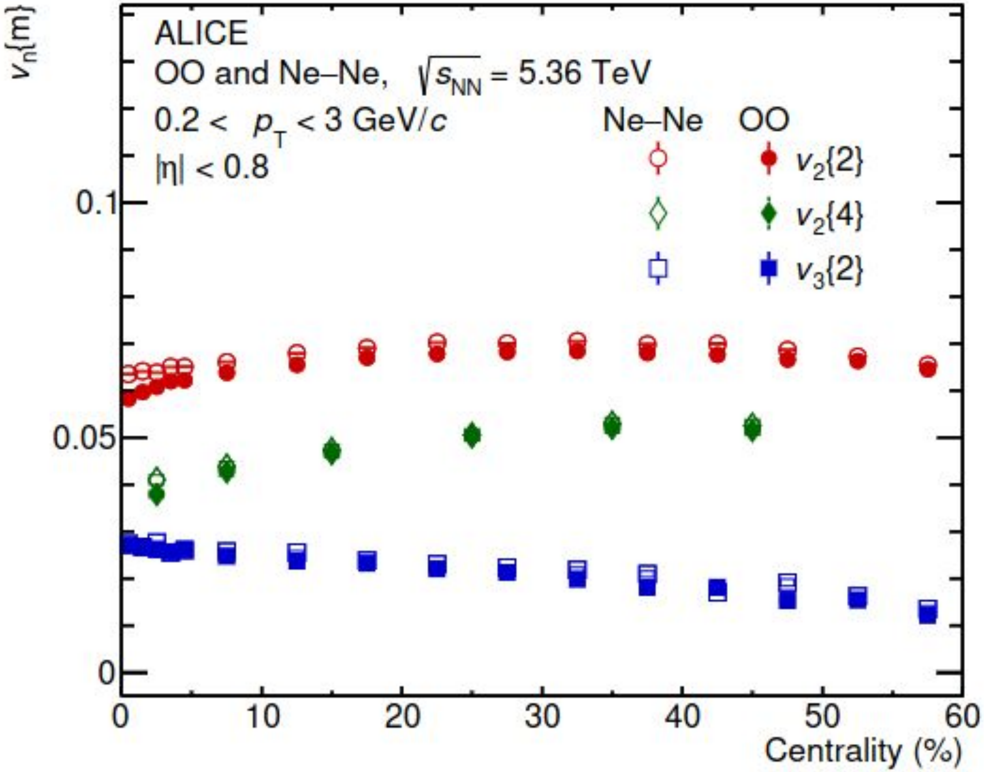
If v_n ~ 0, then no preferred direction
(No flow)

If v_n > 0, then azimuthal correlation

Key Observations:

- Clear centrality dependence
- Non-zero in both systems

-> Flow like correlations



System Comparison

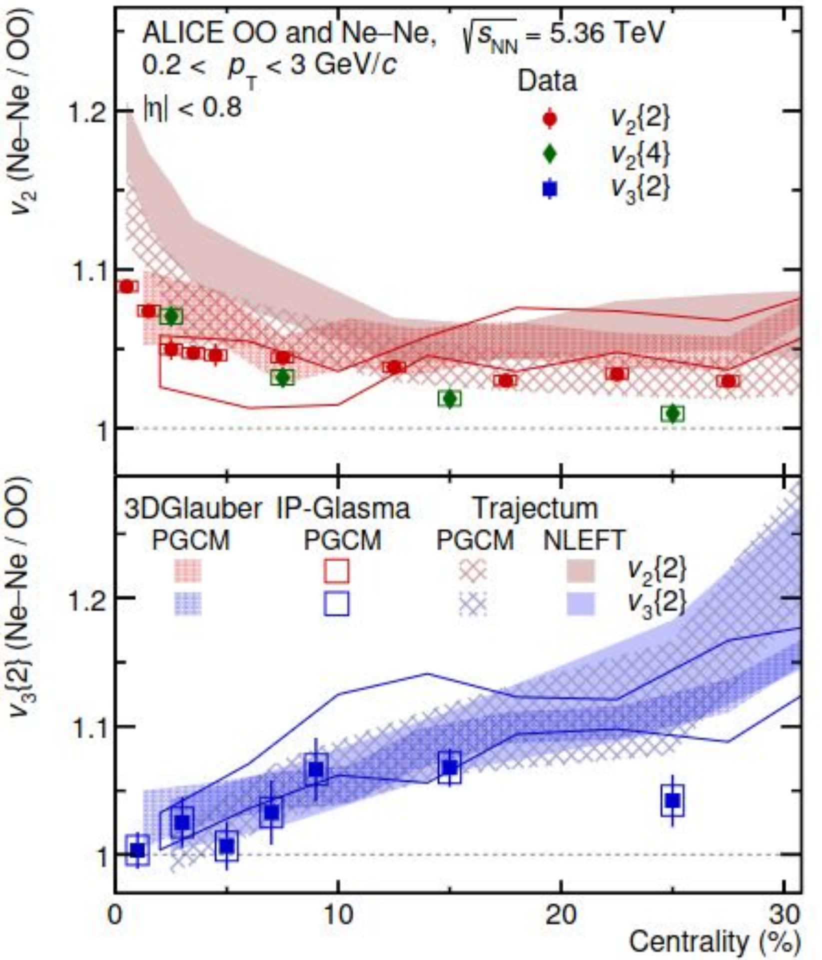
Central coll. = more symmetric overlap

Enhancement is evidence of deformation: Ne-Ne has more quadrupole deformation.

Trend reflects that v_3 is dominated by fluctuations and O-O tetrahedron enhances triangularity.

Observables are sensitive to geometry!

$$v_n \sim \kappa_n \varepsilon_n$$



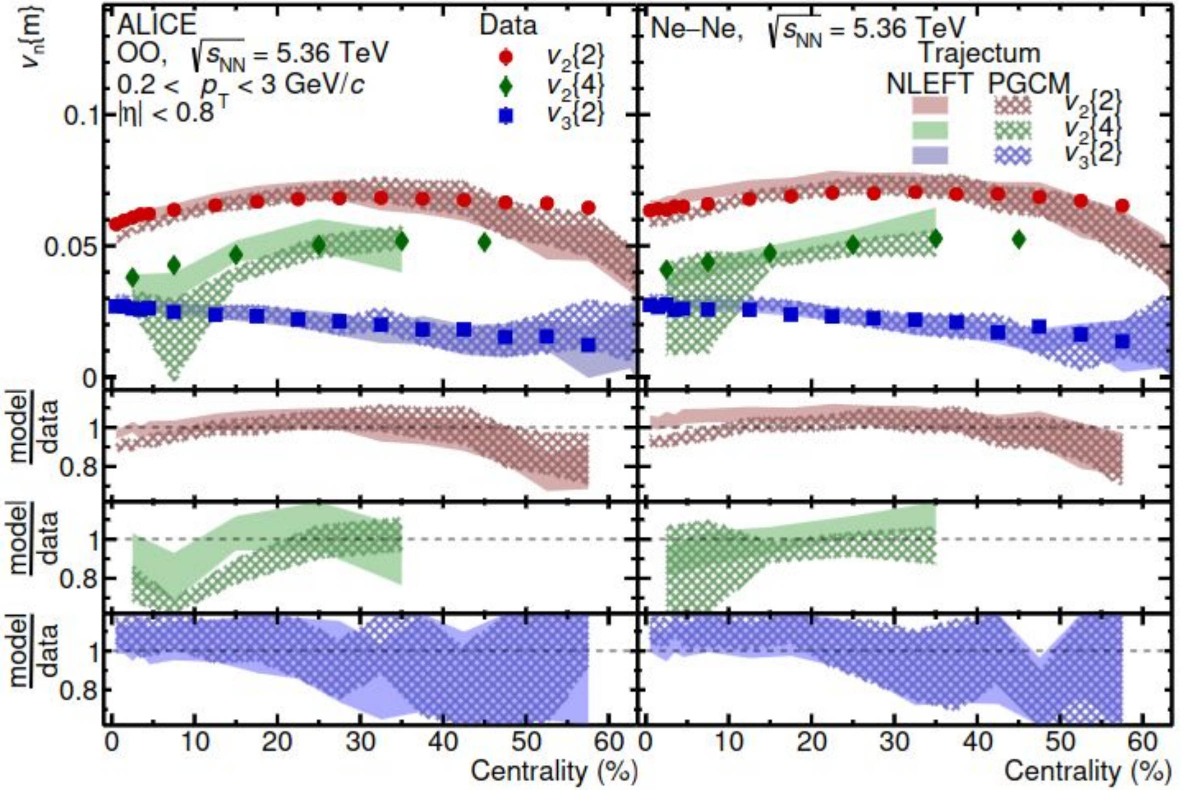
Model Comparisons

Do the current models produce the same signals?

Agreement seen in general centrality dependence.

- Geometry driven trends
- Not a random effect
- Expected hydro response

Cannot simultaneously reproduce all cumulants



Conclusion / Summary

Flow ratios in light-ion collisions provide a geometry-sensitive probe of the nuclear structure by calculating flow coefficients and their ratios, and comparing to theoretical models.

- v_2 increases toward peripheral collisions
- v_3 decreases with centrality
- Ne–Ne / O–O ratios deviate from unity
- Models reproduce trends but fail at cumulant consistency